OR-CAL CAL-SUL 2 Soil Amendment

BETTER YIELDS THROUGH SCIENCE

Safety Data Sheet

Revision Date: July 3rd, 2023

1. Identification

1.1 Product Name: OR-CAL CAL-SUL 2 Soil Amendment

1.2 Other Identification:

Chemical Family: Calcium Polysulfide
Formula: CaSx
CAS Number: 1344-81-6
Product: 28% Calcium Polysulfide

1.3 Recommended Uses:

Soil Amendment

1.4 Manufacturer:

OR-CAL Inc.
29454 Meadowview Rd.
Junction City, OR 97448
541-689-4413 (Office)
541-689-5026 (FAX)
www.orcalinc.com
EPA Establishment No. 52251-OR-005

1.5 Emergency Contact:

CHEMTREC
1-800-424-9300 (US and Canada)
National Pesticide Information Center:
1-800-858-7378
American Association of Poison Control Centers:
1-800-222-1222

2. Hazards

2.1 Hazard Classification:

Health:

Acute Oral Toxicity	Category 4
Acute Dermal Toxicity	Category 4
Acute Inhalation Toxicity	Category 4
Skin Corrosion/Irritation	Category 2
Eye Damage/Irritation	Category 2A

GHS Hazard Categories:
1(Severe), 2/A (Serious), 3(Moderate), 4(Slight), 5(Minimal)

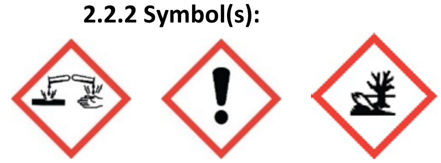
Physical: Corrosive to copper and aluminum

2.2 Signal Word(s): WARNING (GHS), CAUTION (CA)

2.2.1 Hazard Statement(s):

Harmful if swallowed
Harmful in contact with skin
Harmful if inhaled
Causes skin irritation
Causes serious eye irritation
Corrosive
Harmful to aquatic life

2.2.2 Symbol(s):



2.2.3 Precautionary Statement:

- Wash hands thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco, or using the toilet.

- Do not eat, drink or smoke when using this product.
- Wear protective gloves, eye protection, and protective clothing when handling this product. See section 8 for complete list of PPE.
- Take off contaminated clothing and wash it before reuse.
- As soon as possible, wash thoroughly and change into clean clothing after handling or using this product.
- Have the product label or container with you when calling the poison control center or doctor, or going for treatment.
- Avoid breathing mist and use only outdoors in a well-ventilated area.
- Do not allow release to aquatic waterways.

2.3 Unclassified Hazard(s): Potential aquatic toxicity

2.4 Unknown Toxicity Ingredient: Do not mix with acids or phosphate fertilizer products. Deadly and potentially extremely flammable hydrogen sulfide gas may be emitted.

3. Composition/Information on Ingredient

3.1 Product Composition:

Chemical Name	Formula	CAS No.	WT % A.I.	EINECS No.
Calcium Polysulfide	CaSx	1344-81-6	28%	215-709-2
Reaming Components	Trade secret	NA	72%	NA

4. First Aid Measures

4.1 Symptoms/Effects:

Acute: Extremely toxic if swallowed. Decomposition occurs in the digestive tract releasing hydrogen sulfide gas. Causes skin irritation and may produce systemic toxicity by skin absorption. Corrosive to eyes and causes eye irritation and damage. Solution causes alkaline burns

Chronic: No known chronic effects

4.2 Eyes:

If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists, get medical attention.

4.3 Skin:

If on skin: Wash with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice. Take off contaminated clothing and wash it before reuse.

4.4 Ingestion:

If swallowed: Call a poison control center or doctor immediately for treatment advice. Have person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so by the poison control center or doctor. Do not give anything by mouth to an unconscious person.

4.5 Inhalation:

If inhaled: Move person to fresh air and keep person comfortable for breathing. Call a poison control center or doctor for further treatment advice.

5. Firefighting Measures

5.1 Flammable properties: (See section 9 for additional properties)

Do not introduce acids into a vessel containing calcium polysulfide solution. Acids may form highly toxic and extremely flammable hydrogen sulfide gas.

5.2 Extinguishing Media:

5.2.1 Suitable Extinguishing Media:

Water, Foam, Carbon Dioxide, Dry Chemical

5.2.2 Unsuitable Extinguishing Media:

Not Applicable

5.3 Protection of Firefighters:

5.3.1 Specific Hazards Arising from the Chemical

Physical Hazards:

Heating (flames) of closed or sealed containers may cause violent rupture of container due to thermal expansion of compressed gases.

Chemical Hazards:

Heating causes release of hydrogen sulfide vapors. Hydrogen sulfide is a highly flammable gas and gas/air mixtures can be explosive. It may travel to sources of ignition and flash back. Vapors are irritating to eyes, nose, throat and respiratory tract.

5.3.2 Protective Equipment and Precautions for Firefighters:

Wear NIOSH approved self-contained breathing apparatus due to the possible presence of gases and the irritating nature of the product.

5.4 Additional Information:

National Fire Protection Association (NFPA) Ratings:

Health	2
Flammability	0
Instability	0
Special	N/A

Hazard Rating: 4 (severe), 3 (serious), 2 (moderate), 1 (slight), 0 (minimal.)

6. Accidental Release Measures

6.1 Personal Precautions:

Use Personal Protective Equipment specific in section 8. Remove contaminated clothing and wash affected skin areas with soap and water.

6.2 Environmental Precautions:

Keep spill out of open bodies of water and municipal sewers unless allowed under NPDES (National Pollutant Discharge Elimination System) permit. Avoid release to the aquatic environment because of potential aquatic toxicity.

6.3 Methods of Containment

Small Release:

Absorb spillage onto sand, earth or any suitable absorbent material. Shovel up absorbed material and place in drums for disposal as a chemical waste.

Large Release:

Shut off release if safe to do so. Dike spill area with earth, sand or other inert absorbent material to prevent runoff into surface waterways (potential aquatic toxicity).

6.4 Method for Cleanup:

Small Release:

For small areas shovel up absorbed material and place in drums for disposal as a chemical waste.

Large Release:

Dike and contain the spill with neutral or alkaline material (e.g. sand, earth, etc.). Transfer as much liquid as possible to containers for recovery using portable pumps and hoses. Transfer contaminated diking material to treatment facility making sure to follow NPDES permit requirements.

7. Handling and Storage

7.1 Handling:

Avoid prolonged or repeated contact with the skin. Avoid contact with eyes. See section 8 for a complete list of PPE to wear when handling this product.

7.2 Storage:

Store product in a secure locked place, inaccessible to children, pets, and livestock. Make sure the location is cool and dry. Keep container tightly closed with no exposure to air when not in use. Store at a maximum temperature of 110°F and minimum temperature of 7°F.Do not store adjacent to acids.

8. Exposure Control/Personal Protection

8.1 Exposure Guidelines:

8.1.1 OSHA (TWA, STEL):

< 20 ppm of hydrogen sulfide

8.1.2 ACGIH (TLV, STEL):

< 10 ppm of hydrogen sulfide

8.2 Engineering Controls:

Use adequate exhaust ventilation to prevent inhalation of product vapors. Keep eye wash/safety shower in areas where commonly used.

8.3 Personal Protective Equipment (PPE):

8.3.1 Eye/Face Protection:

Goggles or faceshield

8.3.2 Skin Protection:

Coveralls over long-sleeved shirt and long pants, chemical resistant gloves made of any waterproof material, chemical resistant footwear plus socks, chemical resistant apron when mixing, loading or cleaning equipment, and chemical-resistant headgear for overhead exposure.

8.3.3 Respiratory Protection:

NIOSH approved respirator.

8.3.4 Hygiene Considerations:

Wash hands thoroughly after handling this product before eating, drinking, chewing gum, using tobacco, or using the toilet. Change clothing immediately after finished handling this product.

9. Physical and Chemical Properties

9.1 Appearance: Ruby red liquid
9.2 Odor: Rotten eggs
9.3 Odor Threshold: Not Available
9.4 pH: 11-12
9.5 Melting Point/Freezing Point: Not applicable/ 5 °F
9.6 Boiling Point: 215 °F
9.7 Flash Point: Not Available
9.8 Evaporation Rate: (Butyl Acetate=1) Not Available
9.9 Flammability: Not applicable
9.10 Upper/Lower Flammability Limits: Not Applicable
9.11 Vapor Pressure: Not Available
9.12 Vapor Density: (Air=1) Not Available
9.13 Relative Density: 1.250-1.280 (10.6 lbs. per gal.)
9.14 Solubility: Very soluble in water
9.15 Partition Coefficient: Not Available
9.16 Auto-ignition Temperature: >200 °F
9.17 Decomposition Temperature: 165 °F
9.18 Viscosity: 1.76-1.88 cts
9.19 Storage life: At recommended conditions a minimum of 2 years

10. Stability and Reactivity

10.1 Reactivity: Strong oxidizers and acids

10.2 Chemical Stability: Stable if undiluted and not mixed with other chemicals.

10.3 Possibility of Hazardous Reactions: The introduction of acids may form toxic and highly flammable hydrogen sulfide gas. Do not use aluminum and copper fittings as they will corrode.

10.4 Conditions to Avoid: Direct sunlight, excessive heat and freezing conditions.

10.5 Incompatible Materials: Oxidizing agents, acids, zinc, and corrosive to Aluminum and Copper

10.6 Hazardous Decomposition Products: Hydrogen sulfide

11. Toxicological Information

11.1 Oral: Oral Rat (female) LD50: > 920 mg/kg
11.2 Dermal: Dermal Rabbit LD50: >2000 mg/kg.
11.3 Inhalation: INH-Rat LC50: 3.6 mg/l (4 ½ hr. exposure)
11.4 Eye: Corrosive >41.3 Maximum Avg. Score (Wash) in Rabbits
11.5 Chronic/Carcinogenicity: Data not available
11.6 Teratology: Data not available
11.7 Reproduction: Data not available
11.8 Mutagenicity: Data not available
11.9 Other: "…it has been determined that the use of products containing calcium polysulfide as the sole active ingredient would not present a human health hazard to the general public." (USEAP, RED)

12. Ecological Information

12.1 Ecotoxicity:

Green Algae EC 50: 14.1 ppm
Honey Bee LD 50: >25 µg ai/Bee
Avian LD 50: 560 ai/kg
Bobwhite Quail Dietary LC 50: >5000 ai/ppm
Mallard Duck LC 50: >5000 ai/ppm

12.2 Persistence & Degradability: "Calcium Polysulfide present in moist soils and/or on moist foliage is expected to dissociate rapidly; therefore, run-off and erosion into surface water, as present calcium polysulfide, should be negligible." (USEPA, RED)
12.3 Bioaccumulative Potential: This product is not bioaccumulative
12.4 Mobility in Soil: No data available
12.5 Other Adverse Effects: None known at this time
Note: Ecological Information derived from United States Environmental Protection Agency. Reregistration Eligibility Document for Inorganic Polysulfides. 30 September. 2005< http://www.epa.gov/pesticides/reregistration/REDs/inorganic_polysulfides_red.pdf .>

13. Disposal Considerations

Waste material must be disposed of in accordance with the national and local regulations. Do not contaminate ponds, waterways or ditches with chemical or used container. Consult local, county, state, or Federal agencies for acceptable disposal regulations for unused product. Consult local, county, state, or federal agencies for acceptable disposal of product container.

Other Information

Use of Substance/Preparation:

16.1.1 OR-CAL CAL-SUL 2 Soil Amendment is used as a soil amendment for alkaline soil correction and improvement of water penetration. OR-CAL CAL-SUL 2 can also be used as replanting and pre-planting treatments. It’s not for residential use or application to residential sites. Consult the Label for listed application sites. You must read the entire label before use. The label is the law, noncompliance will result in civil or criminal penalties.

The information compiled in this safety data sheet is believed to be accurate and is given in good faith, but it is the user’s responsibility to determine the suitability of this information for the adoption of necessary safety precautions. No warranty or guarantee, expressed or implied, is made regarding performance, stability or otherwise, of this product, as the manner and conditions of use, handling, storage and other factors may involve other additional safety or performance considerations. No suggestions for use are intended as, and nothing herein shall be constructed as a recommendation to infringe any existing patents or violate International, Federal, State, Tribal, or local laws.

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